MASTER TECHNICAL DOCUMENTATION REPORT

Health360:

Comprehensive Healthcare Risk & Disease Analytics

End-to-End Clinical Risk Assessment, Tableau Dashboard Architecture, and Flask Web Deployment Documentation

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DOCUMENT CONTROL	
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Primary Dataset	smart_healthcare_dataset.csv · 5,000 patient records
Runtime / Embed	Python Flask · http://127.0.0.1:5000 · Tableau JavaScript API v2

COHORT KPIs (VERIFIED AGAINST THE DATASET)

TOTAL PATIENT COHORT	AVG BLOOD PRESSURE
5,000 records · 16 fields · 0 nulls	**134.4 mmHg** Stage-1 band (130–139) in every age bracket
AVG GLUCOSE LEVEL	**AVG BODY-MASS INDEX**
134.5 mg/dL above the 100 mg/dL fasting limit	**25.0 kg/m²** 49.5% overweight or obese

1 Executive Summary

Project overview, problem statement and strategic outcomes

Health360 turns a flat file of **5,000 patient records** (16 demographic, vital-sign, lifestyle, symptom and diagnosis fields) into an interactive clinical-risk intelligence platform. Six governed derived fields — age brackets, WHO BMI classes, a composite **Dynamic Risk Tier Index**, its High/Moderate/Low tier, a symptom-burden count and a primary-condition label — feed **nine Tableau worksheets**, **two master dashboards** and a **four-scene executive story**. The story is embedded with the **Tableau JavaScript API v2** inside a Python **Flask** web application (served locally at http://127.0.0.1:5000), so clinicians and hospital managers reach the same analytics from a single browser page.

54.9%	0	28.7%	49.5%
of patients read systolic BP ≥ 130 mmHg	recorded chronic conditions among the 2,625 patients aged ≤ 50	sit in the High-Risk tier (1,434 patients)	are overweight or obese (BMI ≥ 25)

Problem statement — fragmented clinical data silos and undetected early-onset hypertension

Clinical information is normally scattered across silos: vital signs in one report, chronic-disease registers in another, lifestyle and symptom questionnaires in a third. No single view shows *who* is at risk, *why*, and *how much service capacity* that risk will consume. Diagnosis-driven reporting also misses early-onset hypertension. In this cohort, 2,625 patients (52.5%) are aged 50 or younger and **none carries a recorded heart-disease, diabetes or stroke flag**, yet 1,445 of them (55.0%) already read ≥ 130 mmHg systolic and 766 (29.2%) sit in the High-Risk tier. Without an integrated risk view, this silent burden surfaces only when complications are diagnosed.

Key strategic outcomes delivered

- **Data engineering.** A validated 5,000 × 16 dataset (0 nulls, 0 duplicates) and six governed derived fields whose logic was independently re-implemented in pandas; 48 of 48 values visible in the Tableau views reconcile exactly.
- **Dynamic risk stratification.** A transparent additive index (25–100) that separates 1,434 High-Risk (28.7%), 2,771 Moderate (55.4%) and 795 Low-Risk (15.9%) patients.
- **Interactive visual analytics.** Nine worksheets, two dashboards and a four-scene guided story that answer *who is at risk*, *why*, *and how large is the workload*.
- **Web application embedding.** A Flask + Jinja2 + HTML5/CSS3 shell with executive stat cards and a fully interactive story embedded through the Tableau JavaScript API v2.
- **Actionable insight.** Elevated blood pressure is age-flat (52.8–57.0% at ≥ 130 mmHg in every bracket) while recorded chronic disease starts only after age 50; smoking is the strongest behavioural signal; symptom workload is volume-driven. Section 6 converts these into screening, triage and capacity actions.

2 Architecture & Data Engineering Pipeline

From CSV extraction to the browser: system design, feature engineering and the risk-scoring model

2.1 End-to-End System Architecture

Health360 follows a five-stage, loosely coupled architecture (Figure 1). At build time data flows left to right — CSV extraction, Python profiling and validation, Tableau modelling. At run time a thin Flask layer serves an HTML5 front end that embeds the finished story through the Tableau JavaScript API v2. Each stage has one responsibility, so any stage (for example the CSV, by a live clinical feed) can be replaced without touching the others.

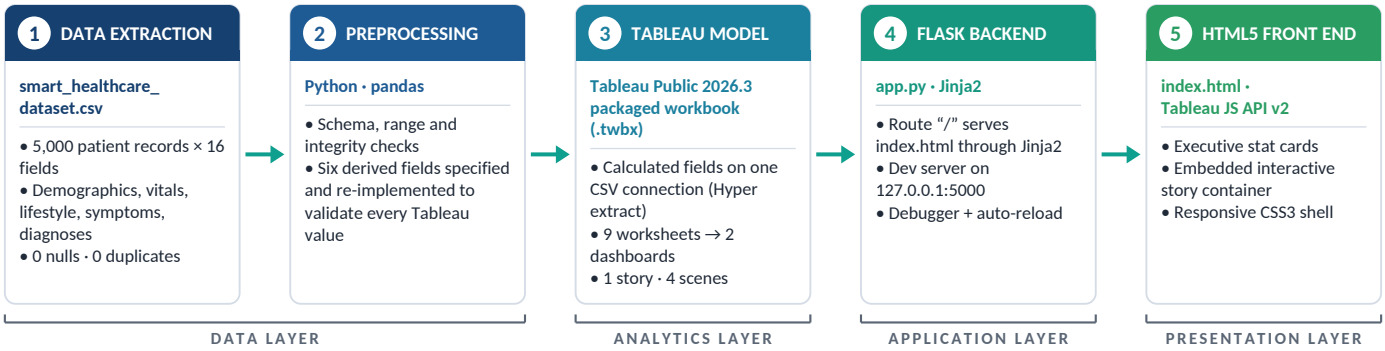


Figure 1 — End-to-end architecture of Health360 — data, analytics, application and presentation layers.

BUILD-TIME PATH	CSV → pandas validation → Tableau model → 9 worksheets → 2 dashboards → **Health360_Executive_Story1**
RUN-TIME PATH	Browser → Flask "/" → Jinja2 `render_template` → HTML5/CSS3 shell → Tableau JS API v2 → interactive story

2.2 Data Preprocessing & Feature Engineering

Six derived fields turn raw columns into analysis-ready dimensions. They are Tableau calculated fields, so the logic travels with the packaged workbook, and were re-implemented independently in Python/pandas to validate every number the dashboards display (Section 6).

Table 1 — Age stratification (4 demographic brackets) and WHO BMI categorisation

Age bracket	Rule (age)	Patients	Share
18–34 · Young Adult	< 35	1,324	26.5%
35–50 · Middle Aged	35 – 50	1,301	26.0%
51–65 · Senior	51 – 65	1,211	24.2%
65+ · Elderly	> 65	1,164	23.3%

BMI category	WHO rule (kg/m²)	Patients	Share
Underweight	< 18.5	516	10.3%
Normal Weight	18.5 – 24.9	2,010	40.2%
Overweight	25.0 – 29.9	1,671	33.4%
Obese	≥ 30.0	803	16.1%

Ages run from 18 to 79; age 65 falls in 'Senior', so 'Elderly' covers 66–79 (label kept as authored in the workbook).

Dynamic Risk Tier Index (DRTI) — formula and tier logic

The source field `health_risk_score` is constant (100.0 in all 5,000 rows), so Health360 engineers a transparent additive index: five measurable inputs earn points from clinically recognised bands and are summed to 25–100, then grouped into three tiers.

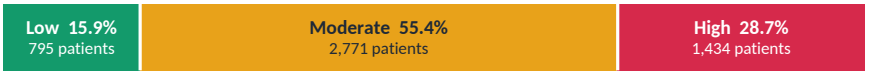
$$\text{DRTI} = P(\text{BP}) + P(\text{Glucose}) + P(\text{Cholesterol}) + P(\text{BMI}) + P(\text{Smoking})$$

Table 2 — Point scores by component

Component (source field)	Lower band	Middle band	Upper band
Systolic blood pressure (mmHg)	≤ 120 → **5** pts	121 – 140 → **15** pts	> 140 → **25** pts
Blood glucose (mg/dL)	≤ 100 → **5** pts	101 – 140 → **15** pts	> 140 → **25** pts
Total cholesterol (mg/dL)	≤ 200 → **10** pts	—	> 200 → **20** pts
Body-mass index (kg/m²)	< 25 → **5** pts	25 – 29.9 → **10** pts	≥ 30 → **20** pts
Smoking (flag)	non-smoker → **0** pts	—	smoker → **10** pts
Range: minimum 5 + 5 + 10 + 5 + 0 = 25 · maximum 25 + 25 + 20 + 20 + 10 = 100 · observed mean 63.7 (Tableau KPI card: 64)			

Worked example (first record in the file): BP 169 → 25 · glucose 74 → 5 · cholesterol 225 → 20 · BMI 22.6 → 5 · smoker → 10; DRTI = **65** → **Moderate Risk**.

- **High Risk** — index ≥ 75
- **Moderate Risk** — index 50 – 74
- **Low Risk** — index < 50



Design note — chronic-disease co-occurrence is analysed, not scored

Heart-disease, diabetes and stroke flags are deliberately not scored, so tiers describe physiological and behavioural risk on the *predictor* side and can be tested against diagnoses instead of being partly defined by them. Because five inputs are scored, their gradients across tiers hold by construction; alcohol, exercise and age are not scored and supply the independent evidence in Sections 3–4.

2.3 Data Dictionary

Table 3 documents the 16 source fields and the 6 derived fields. Ranges, counts and shares were computed from the 5,000-row file.

Field name	Data type	Description	Sample range / values	Transformation applied	Role in analytics
SOURCE FIELDS (16) — smart_healthcare_dataset.csv					
`age`	Integer	Patient age in completed years	18–79; mean 48.8	Binned → Age Brackets	Demographic strata; early-onset test
`gender`	String	Patient sex as recorded	Female 2,506 · Male 2,494	None	Colour split, Disease Prevalence
`bmi`	Float	Body-mass index, kg/m²	15.0–42.6; mean 25.0	Classified → BMI Category; scored	Scatter axis; index input
`exercise_level`	Integer (ordinal)	Activity level coded 0–2 (higher = more active; code labels not supplied)	0: 1,647 · 1: 1,677 · 2: 1,676	None (discrete dimension)	Lifestyle driver
`smoking`	Integer (0/1)	Current-smoker flag	0: 2,462 · 1: 2,538	Scored: +10 points	Lifestyle driver; index input
`alcohol`	Integer (0/1)	Alcohol-consumption flag	0: 2,493 · 1: 2,507	None	Lifestyle driver
`blood_pressure`	Integer	Blood pressure, mmHg (single reading; treated as systolic)	90–179; mean 134.4	Banded and scored (> 120, > 140)	Headline KPI; index input
`cholesterol`	Integer	Total cholesterol, mg/dL	150–299; mean 224.6	Banded and scored (> 200)	Vital indicator; index input
`glucose`	Integer	Blood glucose, mg/dL (fasting status not recorded)	70–199; mean 134.5	Banded and scored (> 100, > 140)	Headline KPI; scatter axis
`fatigue`	Integer (0/1)	Symptom flag — fatigue	1: 2,576 (51.5%)	Summed → Symptom Burden Count	Symptom burden
`chest_pain`	Integer (0/1)	Symptom flag — chest pain	1: 2,520 (50.4%)	Summed → Symptom Burden Count	Symptom burden
`dizziness`	Integer (0/1)	Symptom flag — dizziness	1: 2,487 (49.7%)	Summed → Symptom Burden Count	Symptom burden
`heart_disease`	Integer (0/1)	Heart-disease diagnosis flag	1: 1,499 (30.0%)	Precedence rule → Primary Condition	Outcome / condition
`diabetes`	Integer (0/1)	Diabetes diagnosis flag	1: 1,746 (34.9%)	Precedence rule → Primary Condition	Outcome / condition
`stroke`	Integer (0/1)	Stroke diagnosis flag	1: 1,243 (24.9%)	Precedence rule → Primary Condition	Outcome; colour in Lifestyle Impact
`health_risk_score`	Float	Source-supplied risk score	Constant 100.0 (zero variance)	Excluded; superseded by DRTI	None (non-informative)
DERIVED FIELDS (6) — Tableau calculated fields					
Age Brackets	String	Four age strata	18–34 · 35–50 · 51–65 · 65+	IF / ELSEIF on age (Table 1)	Dimension (worksheet 02)
BMI Category	String	WHO weight class	Underweight · Normal · Overweight · Obese	IF / ELSEIF on bmi (Table 1)	Dimension (worksheet 02)
Dynamic Risk Tier Index	Integer (measure)	Additive composite risk score	25–100; mean 63.7	Sum of five banded point scores (Table 2)	KPI; scatter colour; bar axis
Risk Tier Category	String	High / Moderate / Low tier	≥ 75 · 50–74 · < 50	Thresholds on the index	Dimension (04, 05)
Symptom Burden Count	Integer (measure)	Concurrent symptoms per patient (0–3)	0: 611 · 1: 1,842 · 2: 1,900 · 3: 647	chest_pain + fatigue + dizziness	Heatmap columns (08)
Primary Disease Condition Label	String	Mutually exclusive primary condition	No Chronic 3,215 · HD & Stroke 1,207 · HD 292 · Diabetes 286	Precedence: HD & Stroke > HD > Diabetes > Stroke > none	Dimension (03, 08)

2.4 Profiling Findings & Interpretation Guardrails

- **Integrity.** 5,000 rows × 16 columns; 0 missing values; 0 duplicate rows; every categorical and binary field holds only its expected codes.
- **Zero-variance field.** `health_risk_score` equals 100.0 for every record; it is excluded and superseded by the DRTI.
- **Floor effect.** 132 patients (2.6%) sit exactly at BMI 15.0 — the flat line of marks along the bottom of the Figure 3 scatter.
- **Provenance.** Marginal distributions are close to uniform (age 18–79, BP 90–179, glucose 70–199, cholesterol 150–299), binary flags split ≈ 50/50 and chronic-disease prevalence (heart disease 30.0%, diabetes 34.9%, stroke 24.9%) far exceeds general-population norms. Treat the file as a benchmark cohort: the results demonstrate the analytical method and are not epidemiological estimates.
- **Measurement context.** BP is a single value (treated as systolic) and glucose has no fasting status, so bands act as screening flags, not diagnoses. 'Stage-1' follows the 2017 ACC/AHA systolic band (130–139 mmHg).
- **Hierarchical condition label.** The precedence rule means the 'Diabetes' bar (286) counts diabetics *without* heart disease; the raw diabetes flag is 1,746 (34.9%) and a 'Stroke-only' group never occurs because all 1,243 stroke patients also have diabetes.
- **Association, not causation.** The data are cross-sectional; Section 6 proposes screening and capacity actions, not causal claims.

3 Tableau Visualization & Dashboard Architecture

Nine worksheets, two master dashboards and one story built in Tableau Public 2026.3 from a single data connection

3.1 Overview of the Visual Worksheets

All nine worksheets read one connection to smart_healthcare_dataset.csv (a Hyper extract inside the packaged workbook) and share the six derived fields. Names are numbered to mirror the reading order; blue/orange marks categorical splits, red–green or red–blue ramps mark risk intensity, and counts or means are printed directly on the marks.

Table 4 — The nine analytical worksheets

Worksheet	Chart type	Dimensions	Measures	Clinical insight derived
01_KPI_Cards	KPI text cards	Measure Names	AVG(BP) · COUNT · AVG(Index) · AVG(Glucose)	Four headline numbers — 134 mmHg (134.4), 5,000 patients, index 64 (63.7), 134 mg/dL (134.5) — set the cohort baseline.
02_Age_BMI_Distribution	Highlight table	BMI Category × Age Brackets	COUNT(records): colour + label	Normal Weight leads every bracket (552 / 524 / 485 / 449), yet 48.2% of 18–34-year-olds are overweight or obese (50.9% at 65+): weight risk starts early.
03_Disease_Prevalence	Stacked bar	Primary Condition · Gender	COUNT(records)	3,215 (64.3%) have no chronic condition; Heart Disease & Stroke is the largest chronic segment (1,207; 24.1%); gender split is near even.
04_Risk_Population_Matrix	Text / size matrix	Risk Tier · Exercise Level (aggregated)	COUNT(records): size + label	Sizes the tiers — High 1,434 (28.7%), Moderate 2,771 (55.4%), Low 795 (15.9%); 84.1% of patients sit above the Low tier.
05_Vital_Indicators	Grouped bars	Risk Tier · Measure Names	AVG(BP) · AVG(Chol.) · AVG(Glucose)	High-Risk patients average 154.4 mmHg, 239.6 mg/dL cholesterol, 157.8 mg/dL glucose vs 113.1 / 203.0 / 103.2 for Low Risk (follows from the scoring).
06_Clinical_Risk_Matrix	Scatter (≈ 5,000 marks)	Glucose × BMI · Risk Tier	Index (colour)	Colour shifts at the 100 and 140 mg/dL cut-offs (mean index 51.6 → 61.7 → 71.6); BP and glucose drive the index most (r = 0.57, 0.51; BMI 0.28).
07_Lifestyle_Impact	Bars, 2 × 2 panels	Alcohol × Smoking	AVG(Index) height; SUM(Stroke) colour	Smokers average 68.4 vs 58.9 (≈ +10, as scored); alcohol shifts the index by only 0.2 (63.8 vs 63.6); stroke counts stay at 299–325 per panel.
08_Symptom_Heatmap	Heatmap	Condition × Symptom Count (0–3)	COUNT(records)	Heart Disease & Stroke supplies 629 of 917 chronic patients with ≥ 2 symptoms (68.6%); 1,630 symptomatic patients (32.6%) have no recorded condition.
09_Exercise_vs_Vitals	Bar chart	Exercise Level (0 / 1 / 2)	AVG(Blood Pressure)	Mean BP is 133.8 / 135.4 / 134.1 mmHg at levels 0 / 1 / 2 — indistinguishable (ANOVA p = 0.16); exercise alone does not explain BP.

3.2 Executive Dashboard 1 — Population Vitals & Demographic Breakdown

01_Executive_Health_Overview uses tiled containers with automatic sizing and three horizontal bands. Band 1 is the full-width KPI strip (01). Band 2 pairs the age-by-BMI highlight table (02) with the primary-condition prevalence bars (03). Band 3 pairs the risk-tier matrix (04) with the vital-indicator bars (05). Four legends — record-count scale 122–552, Measure Names, Gender and a size scale (795–2,771) — are docked in a gutter between the left and right panels so that every panel shares one visual vocabulary.

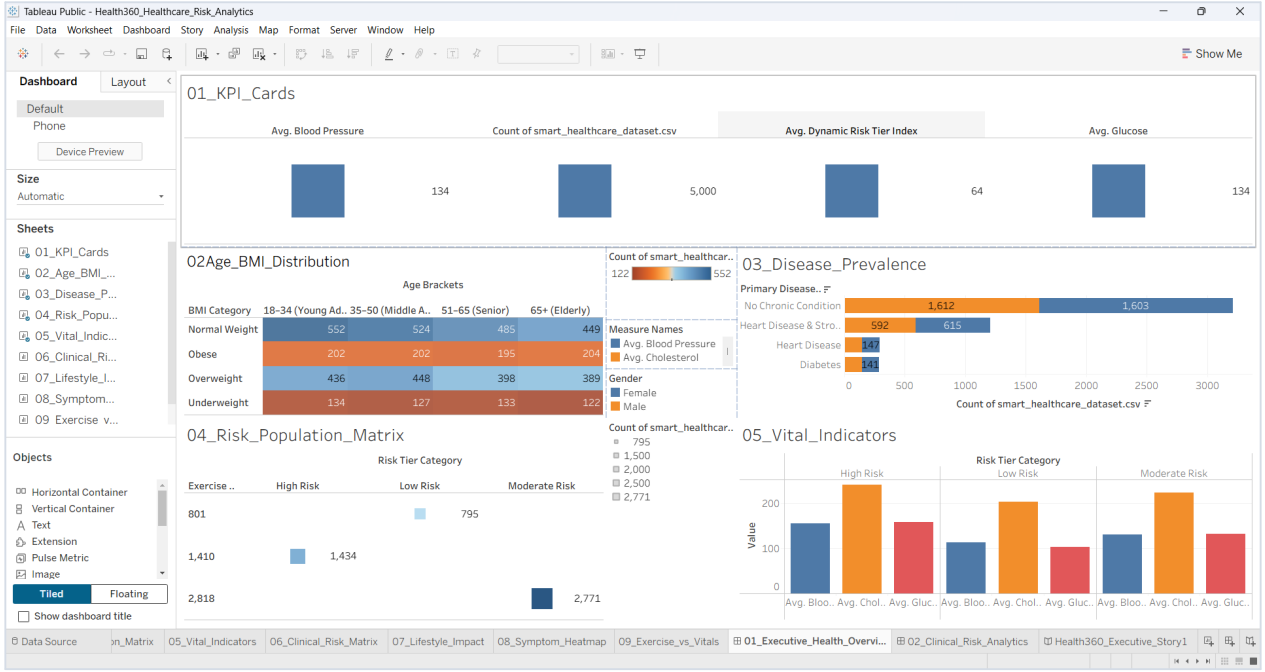


Figure 2 — Executive Overview dashboard (01_Executive_Health_Overview) in Tableau Public: KPI summary cards, age-bracket × BMI table, primary-condition prevalence by gender, risk-tier matrix and vital-indicator bars.

Dashboard 1 — reading the KPI cards. Average systolic BP is **134 mmHg** (134.4 unrounded) — inside the 130–139 mmHg Stage-1 band of the 2017 ACC/AHA classification — and the bracket means (133.7–135.7 mmHg) show a whole-cohort baseline rather than an older-patient effect. Average glucose is **134 mg/dL** (134.5), above the 100 mg/dL fasting upper limit and the 126 mg/dL diabetes threshold, although fasting status is not recorded. The index averages **64** (63.7): the typical patient sits in the Moderate tier, one band below High (≥ 75). The spread behind the BP mean is wide — 34.2% of readings are below 120 mmHg while 44.5% are 140 mmHg or higher — so the cohort is a mix of low- and high-pressure patients, not a uniformly borderline population.

3.3 Executive Dashboard 2 — Clinical Risk Factors & Lifestyle Drivers

`02_Clinical_Risk_Analytics` is a 2 × 2 analytical grid. The top row holds the glucose-by-BMI risk scatter (06, ≈ 5,000 marks coloured by the index) beside the alcohol × smoking bars (07). The bottom row holds the condition-by-symptom heatmap (08) beside mean BP by exercise level (09). Four stacked legends — Dynamic Risk Tier Index (25–250), Stroke (299–325), record count (29–1K) and Exercise Level (0–3K) — sit in the gutter between the columns.

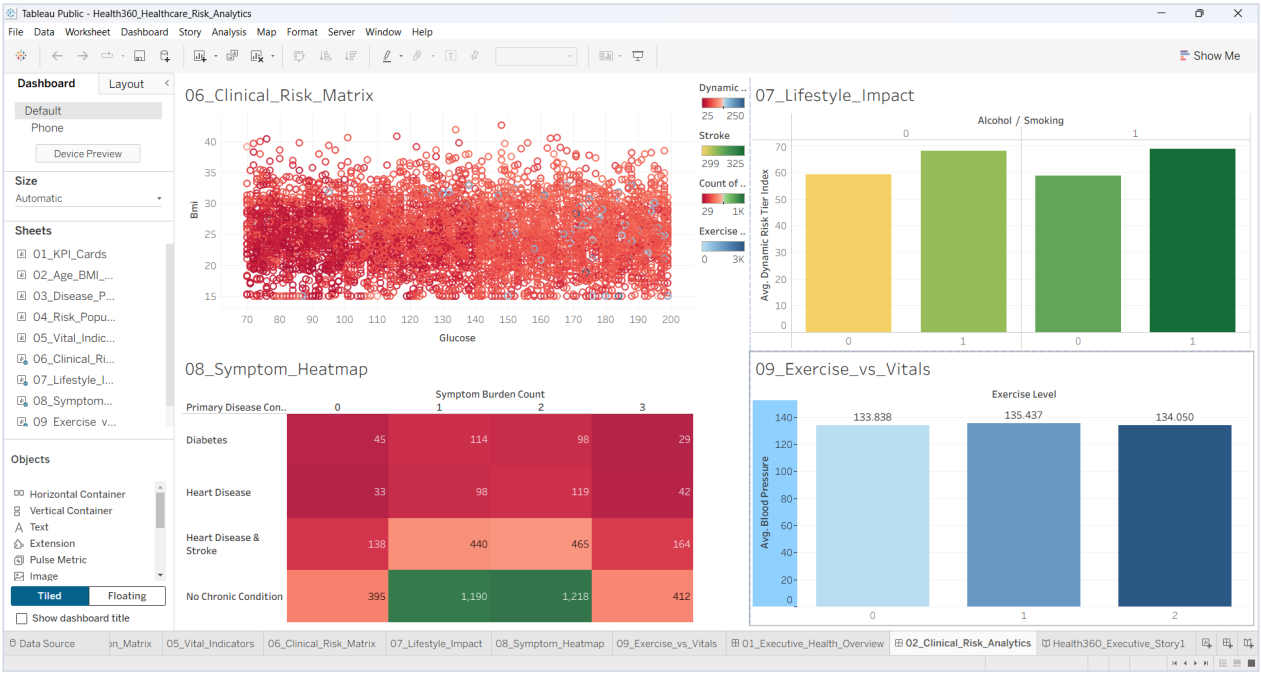


Figure 3 — Clinical Risk Analytics dashboard (02_Clinical_Risk_Analytics): glucose-by-BMI risk scatter, lifestyle-impact bars, symptom-burden heatmap and exercise-versus-BP bars.

Reading the panels. The scatter (06) is the clinical risk surface: colour shifts at the 100 and 140 mg/dL cut-offs of the scoring model (mean index 51.6 → 61.7 → 71.6 across the three glucose bands), BP and glucose drive the index most (r = 0.57 and 0.51; BMI 0.28), and the flat line of marks at BMI 15.0 exposes a floor in the source data (132 patients). In the heatmap (08) Heart Disease & Stroke is the densest chronic row (440 and 465 patients with one or two symptoms) while No Chronic Condition is largest overall (1,190 and 1,218). Panel 09 shows mean BP flat across exercise levels (133.8 / 135.4 / 134.1 mmHg).

How smoking, alcohol and low exercise relate to disease risk

Factor	What Dashboard 2 shows	Evidence	Interpretation
Smoking (scored, +10)	Mean index 68.4 vs 58.9 (Δ 9.5); High-Risk share 39.3% vs 17.7% (2.2×).	Strong — partly by design	Smoking carries +10 points, so part of the gap is mechanical. Outside the score it is corroborated: among patients aged ≥ 51, heart disease is 66.7% vs 59.5% (χ^2 $p < 0.001$); stroke ($p = 0.40$) and diabetes ($p = 0.22$) do not differ.
Alcohol (not scored)	Mean index 63.8 vs 63.6 (Δ 0.2); High-Risk share 28.7% vs 28.6%.	None detected	No separation in vitals ($p = 0.59$), tier share ($p = 0.98$) or heart disease in patients ≥ 51 ($p = 0.66$): monitor, but do not prioritise on this dataset.
Low exercise (not scored)	High-Risk share 28.7% / 30.6% / 26.7% and mean BP 133.8 / 135.4 / 134.1 mmHg at levels 0 / 1 / 2.	Weak	The most active group has the lowest High-Risk share ($p = 0.04$) and, in patients ≥ 51, less heart disease (60.2% vs 64.6%, $p = 0.04$), but there is no dose-response and BP does not differ (ANOVA $p = 0.16$).
Stacked behaviours (smoker + drinker + sedentary)	With 0 / 1 / 2 / 3 adverse behaviours the index rises 59.1 → 62.7 → 65.9 → 68.9 and the High-Risk share 18.3% → 26.4% → 33.6% → 40.7%.	Composite only	The vitals-only sub-score stays flat (59.1 / 58.7 / 58.2 / 58.9): behaviours compound the composite through the smoking weight, not through measurable vital-sign deterioration.

Bottom line for lifestyle drivers

Smoking is the only lifestyle factor with consistent evidence in this cohort; alcohol and low exercise do not measurably worsen vitals or tier. Prevention effort should be weighted accordingly, bearing in mind that behaviours are recorded only as simple flags or three activity levels (exercise_level = 0 is treated as 'lack of exercise').

Interactivity in v1.0. The dashboards rely on native Tableau behaviour — hover tooltips, legend highlighting and mark selection — plus story-point navigation. No custom filter or highlight actions are configured, and the Phone layout is Tableau's auto-generated default; cross-filter actions and tuned phone layouts are logged in the roadmap (Section 6.3).

4 Guided Executive Story & Clinical Insights

Four story points that move from population baseline to risk, drivers and hospital workload

4.1 Executive Tableau Story Architecture

`Health360_Executive_Story1` holds four numbered story points. Each captures a worksheet or dashboard state under a caption tab, so a viewer moves from baseline to risk, drivers and workload with the previous/next arrows or by clicking a tab.

Scenes 1 and 3 capture the two master dashboards; scenes 2 and 4 isolate one worksheet each (05_Vital_Indicators, 08_Symptom_Heatmap) so that each carries a single message.

The web shell embeds the whole container as one frame (Figure 7); the capture shows scene 03 active.

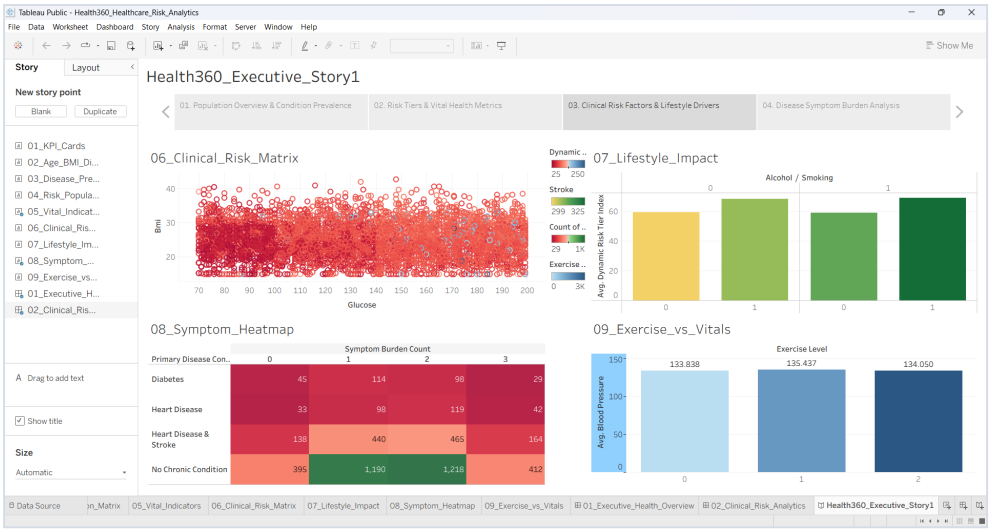


Figure 4 — The four-scene executive story in Tableau Public, scene 03 selected.

4.2 Scene-by-Scene Analytical Walkthrough

Scene (story-point caption)	Captured view	Analytical focus and evidence
1 · Population Overview & Condition Prevalence	Dashboard `01_Executive_Health_Overview`	**Baseline prevalence.** The 5,000-patient cohort (50.1% female), its age and BMI mix (49.5% overweight or obese) and its primary-condition profile: 64.3% no chronic condition, 24.1% Heart Disease & Stroke, 5.8% Heart Disease, 5.7% Diabetes (hierarchical label).
2 · Risk Tiers & Vital Health Metrics	Worksheet `05_Vital_Indicators`	**Tier stratification and vital abnormalities.** High-Risk patients (1,434) average 154.4 mmHg, 239.6 mg/dL cholesterol and 157.8 mg/dL glucose vs 113.1 / 203.0 / 103.2 for Low Risk. Those gradients follow from the scoring; exercise (mean level 0.98 / 1.02 / 1.01) and alcohol (50.2% / 50.3% / 49.3%) do not differ by tier.
3 · Clinical Risk Factors & Lifestyle Drivers	Dashboard `02_Clinical_Risk_Analytics`	**Behavioural drivers and compounding.** Smoking adds ≈ 9.5 index points (39.3% vs 17.7% High-Risk); alcohol adds none; exercise shows no dose-response in BP; stacked adverse behaviours lift the composite index from 59.1 to 68.9, driven by the smoking weight (Table 5).
4 · Disease Symptom Burden Analysis	Worksheet `08_Symptom_Heatmap`	**Symptom burden for hospital resource management.** Heart Disease & Stroke patients account for 629 of the 917 chronic patients with ≥ 2 concurrent symptoms (68.6%); a larger pool — 1,630 patients (32.6% of the cohort) — is symptomatic with no recorded chronic condition. Per-patient symptom load is similar everywhere (1.39–1.58 of 3), so workload is volume-driven.

4.3 Core Clinical Findings & Risk Factors

1 · Early-onset hypertension is age-flat — and undiagnosed. Systolic BP does not rise with age: bracket means are 134.1 / 134.4 / 133.7 / 135.7 mmHg (ANOVA p = 0.28) and 54.5% of 18–34-year-olds read ≥ 130 mmHg versus 57.0% at 65+; 43.3% of the youngest bracket exceed 140 mmHg. Yet none of the 2,625 patients aged ≤ 50 has a recorded condition (first heart-disease flag at age 52, diabetes 56, stroke 63), so 1,445 elevated readings and 766 High-Risk patients are invisible to diagnosis-driven reporting. A single reading is not a diagnosis: treat this as a screening signal.

2 · Multi-condition escalation after 50. Any recorded condition rises from 0% (≤ 50) to 51.3% (51–65) and 100% (65+). Conditions cluster: 1,207 patients (24.1% of the cohort; 50.8% of those over 50) carry all three flags, versus only 289 with one and 289 with two. Every stroke patient is also diabetic (1,243 of 1,243) and 97.1% also have heart disease.

3 · Symptom load is volume-driven. Mean symptoms per patient are almost identical across condition groups (1.39–1.58 of 3) and across 0–3 conditions (1.47–1.54), so symptoms do not escalate with disease count. Heart Disease & Stroke dominates absolute chronic workload (68.6% of chronic patients with ≥ 2 symptoms), while the biggest symptomatic pool — 1,630 patients — needs diagnostic work-up rather than chronic-disease management.

Early-onset gap by age bracket

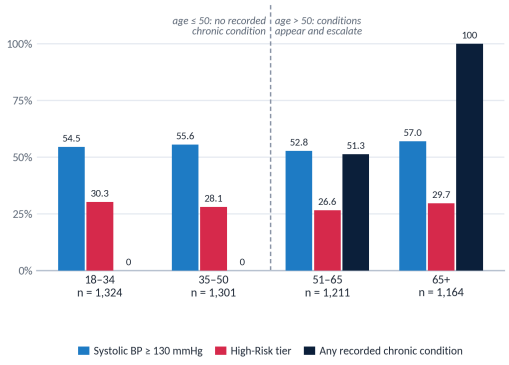


Figure 5 — Elevated BP and High-Risk share are flat across ages; recorded chronic disease appears only after 50 (supplementary analysis computed from the same dataset).

5 Web Integration & Flask Deployment

A lightweight Python/Flask shell that serves the HTML5 front end and embeds the Tableau story

5.1 Technical Web Stack Overview

Layer	Technology	Responsibility in Health360
Server and routing	Python Flask — `app.py`	WSGI micro-framework: defines the "/" route and runs the development server on 127.0.0.1 with the interactive debugger and auto-reload enabled.
Templating	Jinja2 — `render_template`	Loads templates/index.html from the templates folder; the same mechanism lets later versions inject server-side values into the page.
UI shell	HTML5 + CSS3	Platform header with "Tableau JS API v2" and "5,000 Patient Cohort" badges, four colour-coded executive stat cards and a responsive story container.
Visual embed	Tableau JavaScript API v2	Instantiates the published Health360_Executive_Story1 in the container and exposes story-point navigation; embedded views stay fully interactive.
Analytics source	Tableau Public workbook (.twbx)	9 worksheets, 2 dashboards and 1 story with 4 captioned story points — the single source of every visual on the page.

5.2 Codebase Implementation & Directory Structure

Repository layout

```
Health360-Healthcare-Risk-Analytics/
├── app.py                  # Flask entry point
├── templates/
│   └── index.html          # HTML5 shell + embed
├── data/
│   └── smart_healthcare_dataset.csv
└── tableau/
    └── Health360_Healthcare_Risk_Analytics.twbx
```

Run it locally

1. `pip install flask`
2. `python app.py`
3. Open http://127.0.0.1:5000 in a browser.
4. Optional: set the `PORT` variable to change the port (the capture in Figure 6 used 5001).

`app.py` — complete source

```python
"""Health360 - Flask application shell.

Serves the HTML5 front end that embeds the Tableau
executive story through the Tableau JavaScript API v2.
"""
import os
from flask import Flask, render_template

app = Flask(__name__)  # resolves ./templates automatically


@app.route("/")
def home():
    """Render the Health360 executive analytics page."""
    return render_template("index.html")


if __name__ == "__main__":
    port = int(os.environ.get("PORT", 5000))  # default 5000
    # Development server: interactive debugger + auto-reload
    app.run(host="127.0.0.1", port=port, debug=True)
```

5.3 Development Environment Execution

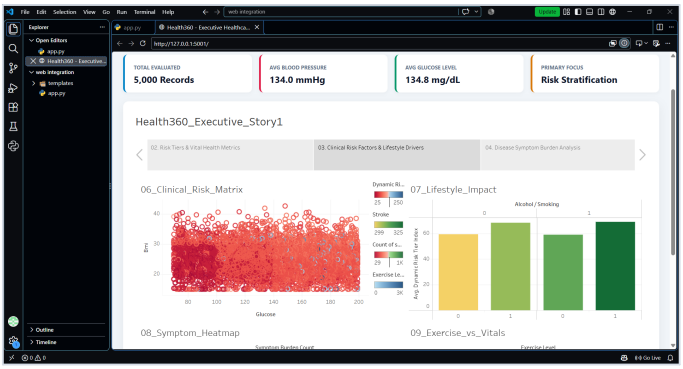


Figure 6 — Development environment: VS Code workspace (Explorer: web integration/ → templates/, app.py) with the Flask-served page open in the integrated browser at 127.0.0.1:5001 (PORT override; default 5000).

What the capture verifies

- The page is served on the loopback interface and renders in the integrated browser.
- Stat cards and the embedded story load; tabs 02–04 are visible with navigation arrows (tab 01 is scrolled out of the narrow pane).
- Run mode: `debug=True` switches on the interactive debugger and auto-reload (development only).

5.4 End-User Web Application Showcase

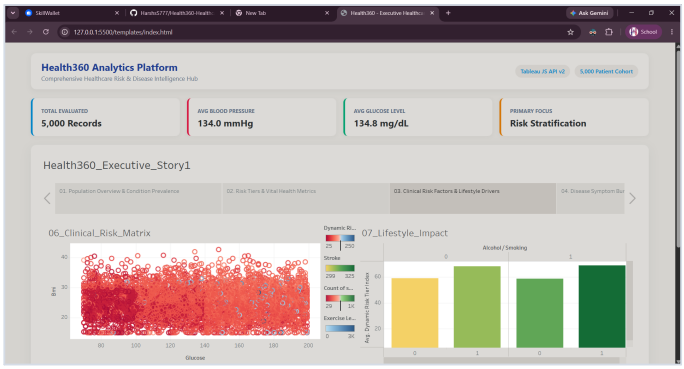


Figure 7 — Chrome showing the live Health360 web interface with the embedded Tableau story (served by VS Code Live Server at 127.0.0.1:5500 for front-end verification).

Features of the web shell

- Executive stat cards: Total Evaluated, Avg Blood Pressure, Avg Glucose Level, Primary Focus.
- Smooth loading animations and a clean card layout around the story.
- Fully interactive embedded frame: four captioned tabs with arrows; scene 03 active here.

Deployment path — from localhost to production

Development: **python app.py** on 127.0.0.1:5000 with debug on. Production: serve the same app with a WSGI server (Waitress on Windows, Gunicorn on Linux) behind an HTTPS reverse proxy, debug off, dependencies pinned in requirements.txt and the port taken from the PORT variable. Tableau Public exposes its data openly, so publish only de-identified data there; real patient data belongs on Tableau Server or Cloud with authentication.

6 Testing, Deployment & Strategic Recommendations

Verification results, evidence-based actions and the maintenance and growth roadmap

6.1 Performance & Usability Testing Report

Testing spans four layers: data integrity, analytical correctness, functional behaviour of the web shell and non-functional targets. Checks T1–T6 were executed for this release; N1–N4 fix the acceptance budgets for timings that must be measured on the deployment machine.

Table 7 — Verification executed (T1–T6) and acceptance targets (N1–N4)

ID	Check / area	Method and scope	Result or acceptance target	Status
EXECUTED FOR RELEASE v1.0				
T1	**Schema and integrity**	pandas checks on the CSV	5,000 × 16; 0 nulls; 0 duplicates; all code domains valid.	✔ **PASS**
T2	**Range plausibility**	Min / max of every vital	Age 18–79, BP 90–179, glucose 70–199, cholesterol 150–299, BMI 15.0–42.6 (132 values at the 15.0 floor).	✔ **PASS**
T3	**Index logic**	Recompute DRTI and tiers	Index 25–100, mean 63.7; tiers 1,434 + 2,771 + 795 = 5,000.	✔ **PASS**
T4	**Derived-field reconciliation**	pandas re-implementation vs values visible in Tableau	48 / 48 identical: age × BMI (16), symptom heatmap (16), condition × gender (6), tiers (3), BP by exercise (3), KPI cards (4).	✔ **PASS**
T5	**KPI consistency across layers**	Dataset vs Tableau vs web shell	Tableau cards match at integer precision; the web shell shows 134.0 mmHg and 134.8 mg/dL against dataset means of 134.4 and 134.5 — update two strings in index.html (v1.1).	⚠ **ATTENTION**
T6	**Functional UI**	Web shell in the VS Code browser (Flask, 5001) and Chrome (Live Server, 5500)	Page renders; 4 stat cards; story with tabs 01–04 and arrows; scene 03 charts render in both (Figures 6–7).	✔ **PASS**
ACCEPTANCE TARGETS — to be recorded on the deployment machine				
N1	**Page-shell load**	Chrome DevTools + Lighthouse; 5 cold-cache runs, median	HTML + CSS interactive ≤ 1 s on localhost; shell payload ≤ 100 KB excluding Tableau assets.	Target
N2	**Embedded-frame response**	First-interactive callback timestamp (performance.now()); stopwatch for navigation	Story first-interactive ≤ 5 s on 10 Mbps; story-point switch ≤ 1.5 s.	Target
N3	**API rendering speed**	DevTools Performance panel and console audit	06_Clinical_Risk_Matrix (≈ 5,000 marks) paints ≤ 2 s; zero console errors.	Target
N4	**Cross-browser compatibility**	Manual smoke test using the T6 checklist	Identical render and navigation in Chrome, Edge, Firefox and Safari — Chrome and the VS Code browser verified; others pending.	Target

Measured footprint: dataset 242.7 KiB, packaged workbook 72.1 KiB; the Flask route renders one template per request and reads no data at run time.

6.2 Strategic Clinical & Hospital Operations Recommendations

Table 9 — Analytical proposals for clinical-governance review (not clinical guidance)

#	Recommendation	Evidence from this cohort	Proposed action and KPI
1	**Early-onset hypertension screening (ages 18–50)**	1,445 of 2,625 patients aged ≤ 50 (55.0%) read ≥ 130 mmHg; 1,145 (43.6%) exceed 140; none has a recorded condition.	BP check at every contact for adults 18–50; confirm raised readings by repeat, home or ambulatory measurement. KPI: share with a confirmed BP status.
2	**Risk-tiered triage protocol**	High 1,434 (28.7%) · Moderate 2,771 (55.4%) · Low 795 (15.9%); 84.1% need more than routine care.	High → fast-track review and metabolic work-up; Moderate → lifestyle programme with scheduled re-scoring; Low → annual screening. KPI: time-to-review.
3	**Confirm undiagnosed hyperglycaemia**	1,175 patients aged ≤ 50 (44.8%) have glucose > 140 mg/dL and no diabetes flag (flags start at age 56); fasting status unknown.	Confirmatory fasting glucose or HbA1c for every reading > 140 mg/dL. KPI: confirmation rate and new-diagnosis yield.
4	**Prioritise smoking cessation**	50.8% smoke; High-Risk share 39.3% vs 17.7% (partly scored); heart disease in patients ≥ 51: 66.7% vs 59.5% ($p < 0.001$).	Embed cessation counselling in High and Moderate pathways. KPI: quit-attempt and 3-month abstinence rates.
5	**Weight management from young adulthood**	49.5% overweight or obese, including 48.2% of 18–34-year-olds; obesity 15.3% (18–34) vs 17.5% (65+).	Offer lifestyle programmes at the first overweight reading, not at diagnosis. KPI: BMI-category migration at 12 months.
6	**Symptom-driven capacity planning**	917 chronic patients carry ≥ 2 symptoms (68.6% Heart Disease & Stroke); 1,630 symptomatic patients (32.6%) have no diagnosis.	Size cardiology/stroke follow-up on Heart Disease & Stroke volume; ring-fence diagnostics (ECG, glucose, lipids) for the symptomatic-undiagnosed. KPI: diagnostic yield, waiting time.

6.3 Maintenance & Future Roadmap

MAINTENANCE PLAN

- **Version control.** Tag releases; keep the .twbx, CSV and code in Git; log threshold changes.
- **Threshold governance.** Review index bands and BP/BMI cut-offs when guidelines change; re-run T1–T4 after any edit.
- **Data refresh.** Re-extract, re-run integrity and reconciliation checks, then republish.
- **Data upgrade.** Replace the constant risk score; capture systolic/diastolic BP, glucose context and outcome dates.
- **Privacy.** De-identify before publishing; HTTPS, authentication and audit logging in production (HIPAA / India's DPDP Act 2023).

FUTURE ROADMAP

- **Phase 2 — real-time patient streaming.** Ingest HL7 FHIR / device vitals through a message queue, refresh extracts incrementally and push live KPIs to the stat cards.
- **Phase 3 — predictive ML via Flask APIs.** Train and calibrate a risk model on recorded outcomes; JSON /api/predict endpoint with SHAP explanations beside the DRTI.
- **Phase 4 — mobile optimisation.** Tuned Tableau Phone layouts, dashboard filter actions, a responsive CSS grid, touch-friendly story navigation and an installable PWA.

Document control — v1.0 · 29 October 2026 · Harshdeep Singh Sisodiya · github.com/HarshsS777/Health360-Healthcare-Risk-Analytics